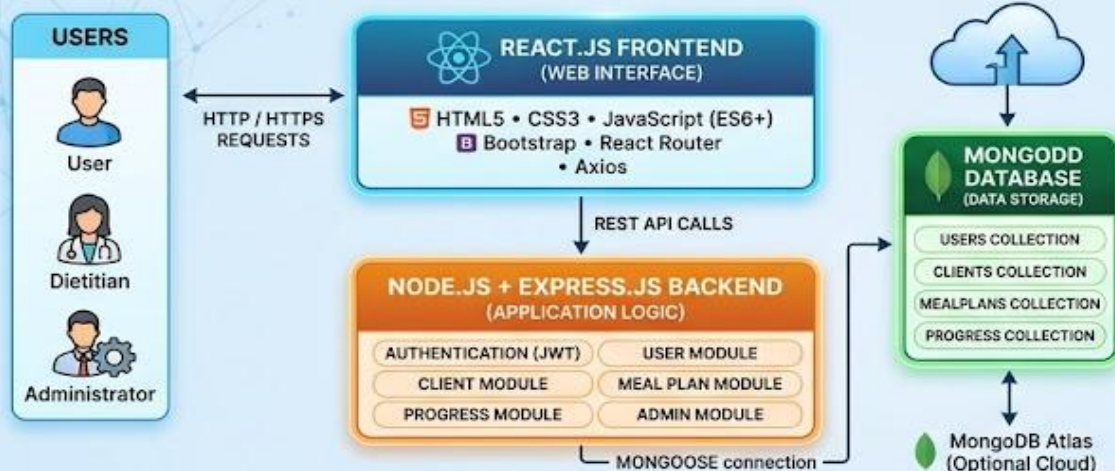


Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	3 rd July,2026
Team ID	
Project Name	NUTRITION ASSISTANT
Maximum Marks	4 Marks

TECHNOLOGY STACK & ARCHITECTURE: NUTRITION ASSISTANT APPLICATION

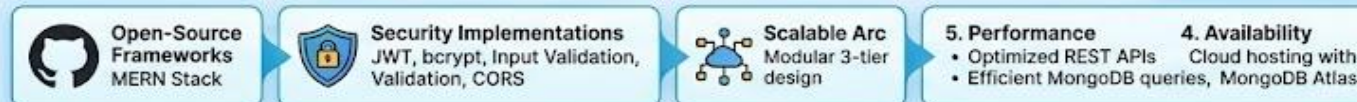
TECHNICAL ARCHITECTURE



KEY SYSTEM COMPONENTS & TECHNOLOGIES



KEY APPLICATION CHARACTERISTICS



TECHNOLOGY SUMMARY

TECHNOLOGY SUMMARY	TECHNOLOGY SUMMARY					
	Frontend	Backend	Database	Authentication & Security	Development Tools	Deployment
	- HTML • Interface - CSS3 • CSS3 - JavaScript (ES6+•t+)	- Frontend: Node.js - Backend: Express.js - Database: MongoDB	Mongoose • Collection	- Authentication (JWT) - Progress Module - Admin Module	- Visual Studio Code - Git - Postman, npm	- Frontend: Vercel - Backend: Render/Railway - Database: MongoDB Atlas

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive web interface for users, dietitians, and administrators	HTML5, CSS3, JavaScript, React.js, Bootstrap
2	Application Logic-1	Authentication, authorization, and business logic	Node.js, Express.js
3	Application Logic-2	Client management, meal planning, nutrient analysis, and progress tracking	Express.js REST APIs
4	Application Logic-3	Database operations and data validation	Mongoose ODM
5	Database	Stores user, client, meal plan, and progress information	MongoDB
6	Cloud Database	Optional cloud-hosted database	MongoDB Atlas
7	File Storage	Profile images or uploaded documents (optional)	Local File System / Cloud Storage
8	External API-1	Nutrition or food database integration (optional)	USDA FoodData Central API / Nutritionix API
9	External API-2	Email notifications for registration or password reset	Nodemailer (SMTP)
10	Machine Learning Model	Personalized meal recommendation (Future Enhancement)	Not Applicable (Current Version)
11	Infrastructure (Server / Cloud)	Application deployment and hosting	Local Server / Render / Railway / Vercel / MongoDB Atlas

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend and backend development	React.js, Node.js, Express.js, MongoDB, Bootstrap, Axios

2	Security Implementations	Secure authentication, password encryption, and role-based access	JWT, bcrypt, Express Middleware, Input Validation, CORS
3	Scalable Architecture	Modular 3-tier architecture supporting future expansion	MERN Stack (Client → Server → Database)
4	Availability	Continuous availability through cloud deployment and reliable database connectivity	MongoDB Atlas, Render/Railway/Vercel (optional deployment)
5	Performance	Optimized REST APIs, efficient MongoDB queries, lazy loading, reusable React components, and code splitting	MongoDB Indexing, React Lazy Loading, Axios, Express.js

Technology Summary

Frontend

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- Bootstrap
- React Router
- Axios

Backend

- Node.js
- Express.js

Database

- MongoDB

- **Mongoose ODM**
- **MongoDB Atlas (Cloud)**

Authentication & Security

- **JSON Web Token (JWT)**
- **bcrypt**
- **Role-Based Access Control**
- **CORS**
- **Input Validation**

Development Tools

- **Visual Studio Code**
- **Git & GitHub**
- **Postman**
- **npm**

Deployment (Optional)

- **Frontend: Vercel**
- **Backend: Render or Railway**
- **Database: MongoDB Atlas**